Serverless data pipeline: API → Lambda → DynamoDB → Lambda → S3

Architecture
A Python Lambda (import-jsonplaceholder-users) fetches 10 users from https://jsonplaceholder.typicode.com/users and writes each one to the DynamoDB table jsonplaceholder-users with PutItem. A second Lambda (export-dynamodb-to-s3) scans the whole table and writes a timestamped JSON file to the private S3 bucket YOUR_UNIQUE_BUCKET_NAME under backups/YYYY-MM-DD/. EventBridge Scheduler invokes the export
Lambda every 2 minutes. Both functions log to CloudWatch.
All resources run in eu-central-1.

Resources
- DynamoDB table: jsonplaceholder-users, partition key id (Number), on-demand
- S3 bucket: YOUR_UNIQUE_BUCKET_NAME (Block Public Access on, SSE-S3 encryption)
- Lambda #1: import-jsonplaceholder-users (Python 3.12, 128 MB, 30 s)
- Lambda #2: export-dynamodb-to-s3 (Python 3.12, 128 MB, 30 s)
- Schedule: dynamodb-backup-every-2min, rate(2 minutes), flexible window off

Results
- Lambda #1 test: statusCode 200, saved_count 10 (screenshot 03, logs 04)
- DynamoDB scan: 10 items (screenshot 05)
- Lambda #2 manual tests: 2 separate JSON files (screenshots 06, 07)
- Scheduled runs: 43 files created automatically about 2 minutes apart (screenshot 09)
- Lambda #2 metrics: 42 invocations, 0 errors, ~490 ms duration (screenshot 10)

IAM (least privilege)
Lambda #1 only writes users, so its role allows only dynamodb:PutItem on this one table. Lambda #2 only reads the table and creates backup files, so its role allows dynamodb:Scan on the table and s3:PutObject on backups/* in this bucket. Neither can delete data. Both keep AWSLambdaBasicExecutionRole for CloudWatch Logs.

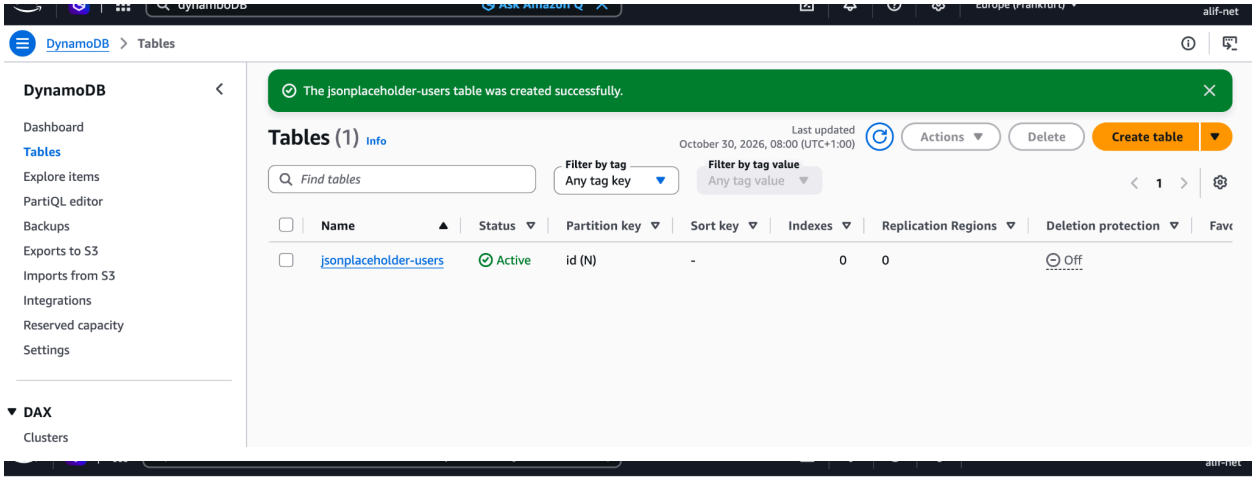
Answers
1. The Lambda execution role is the identity the function uses while it runs: boto3 gets temporary credentials from it to call DynamoDB, S3 and CloudWatch. The Scheduler execution role is the identity EventBridge Scheduler uses to start the function; it only needs lambda:InvokeFunction on that Lambda.
2. PutItem writes by partition key. The second run sends the same ids 1-10, so each existing item is replaced, not added. The table stays at 10 items.
3. The key contains the UTC date, the time down to microseconds and the Lambda request ID, so every run creates a new object instead of overwriting the last one, files are grouped by day, and each file can be traced to its exact log.
4. print() output goes to CloudWatch Logs and describes what happened inside one run (which users were saved, error messages). Invocations and Errors are

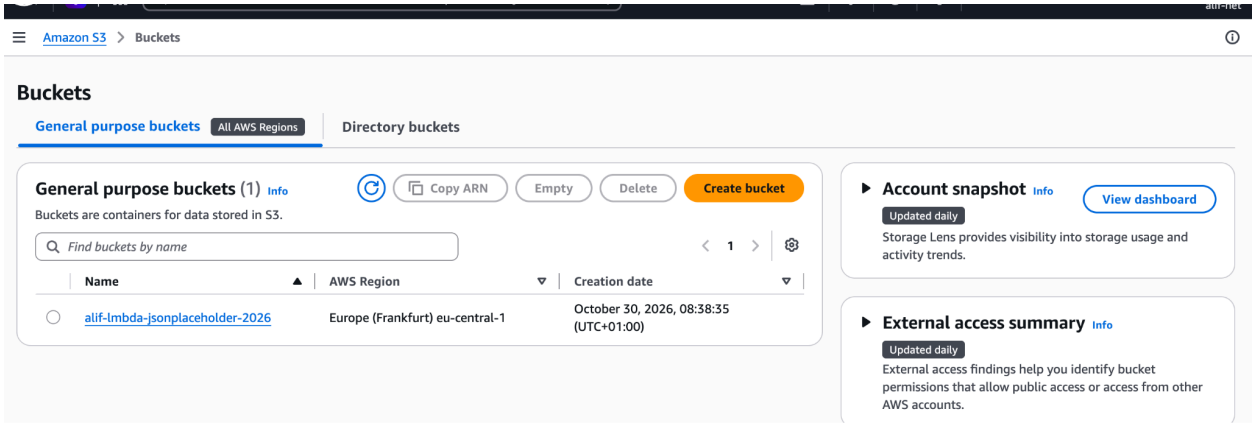
metrics AWS records automatically: counts over time, used for graphs and alarms. Metrics show that something went wrong; logs show what.

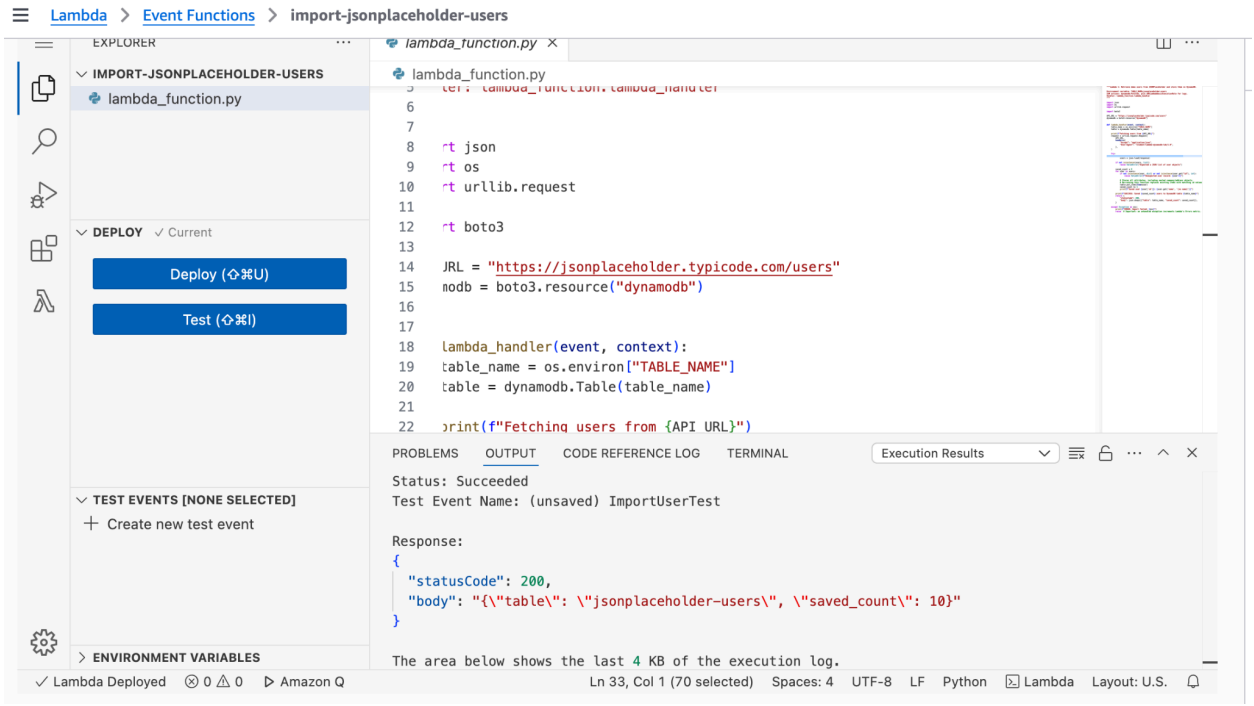
5. The item data would survive in S3, but the table would not come back on its own. The JSON holds only items, not the key schema, capacity mode, indexes, TTL, streams, PITR settings, tags, policies or alarms. I would recreate the table (ideally from infrastructure as code) and re-import the items with a script.
6. Every export scans the whole table: it costs read capacity, gets slower as the table grows, can hit Lambda's 15-minute and memory limits, is not a consistent point-in-time snapshot, and creates about 720 near-duplicate files per day. For production I would enable PITR, use DynamoDB Export to S3 or AWS Backup, and choose frequency and retention from the RPO/RTO, with tested restores.

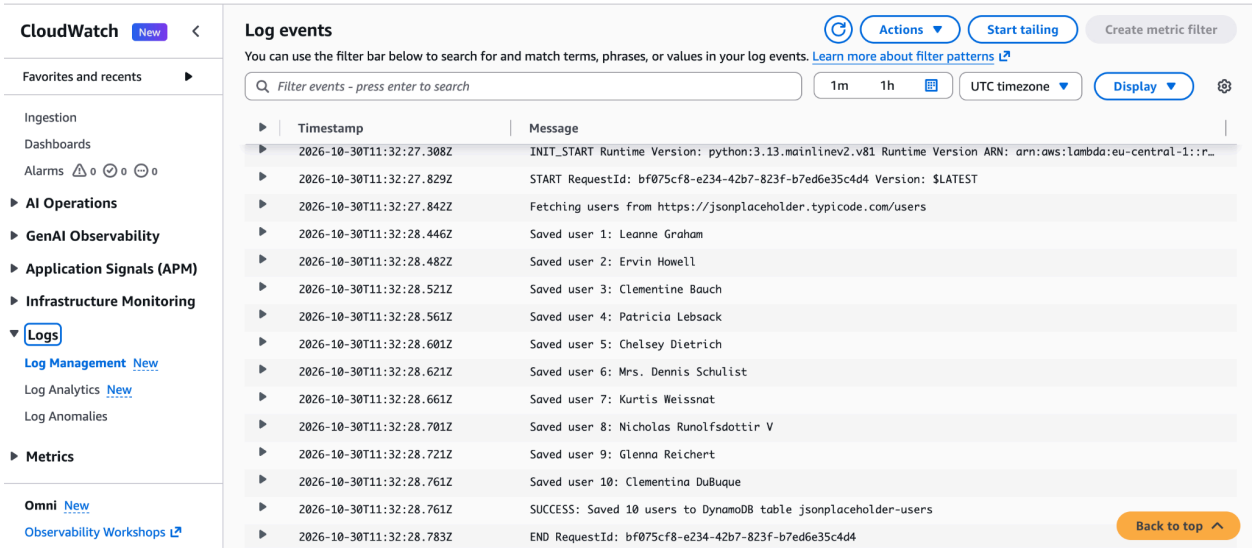
Cleanup

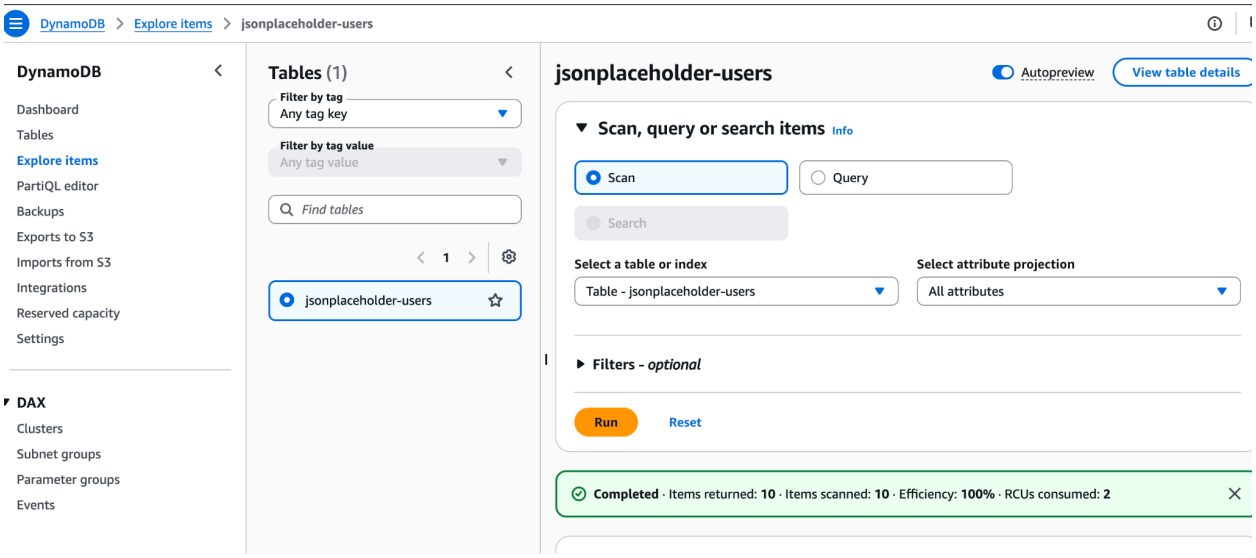
Schedule disabled and deleted, both Lambdas, the table, the bucket, log groups and lab IAM roles deleted (screenshot 12).

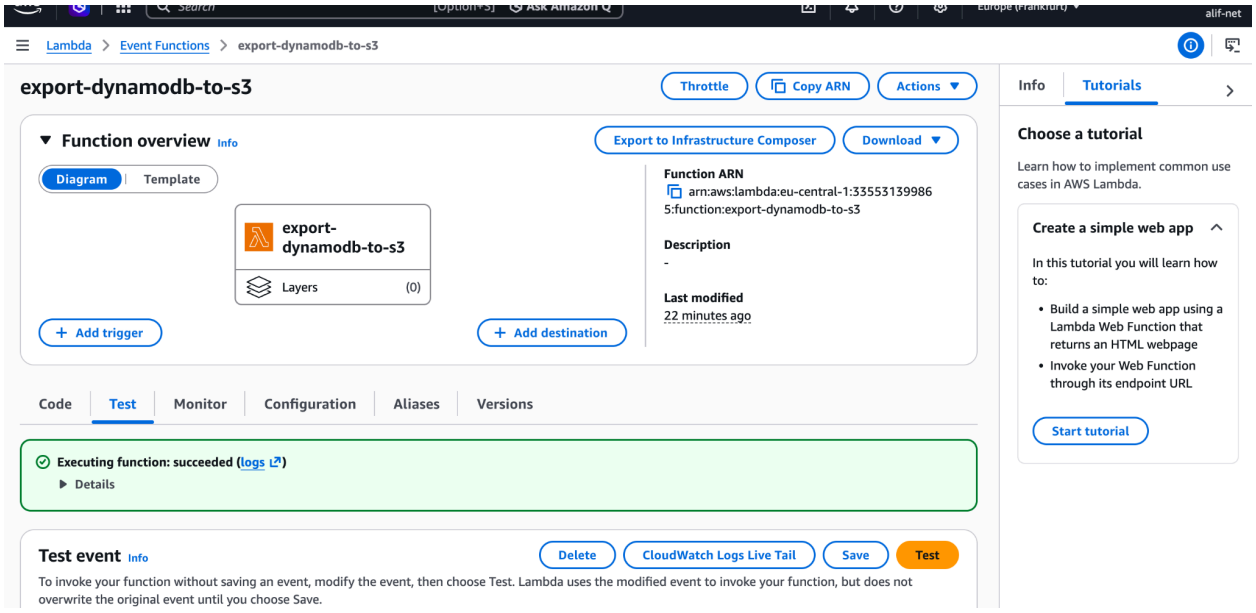


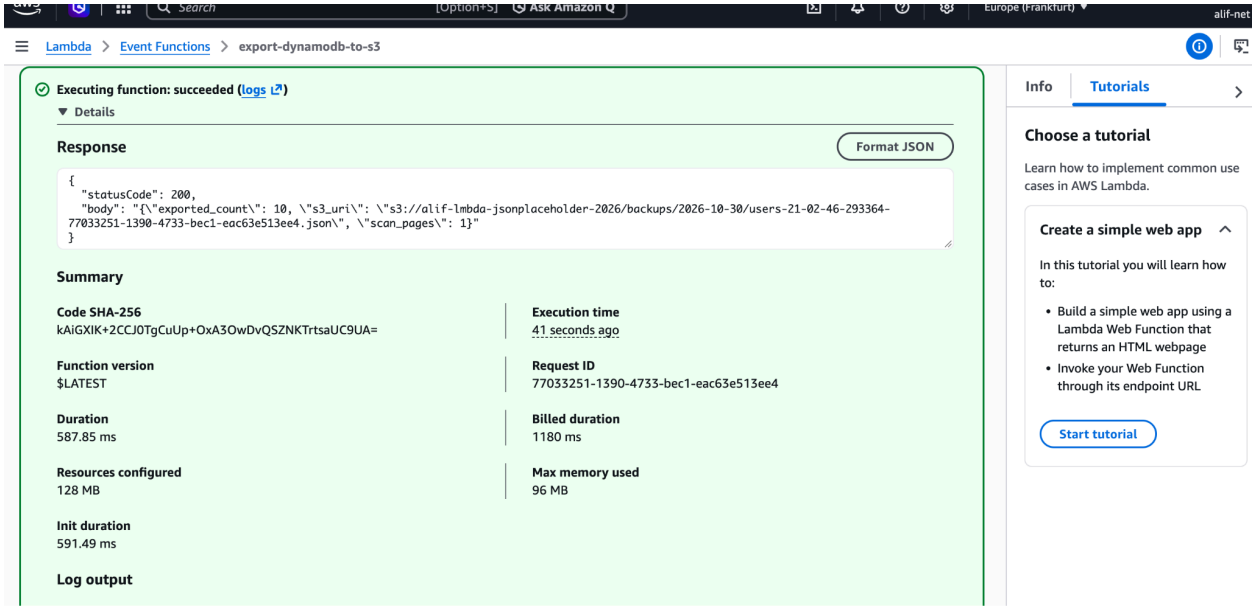


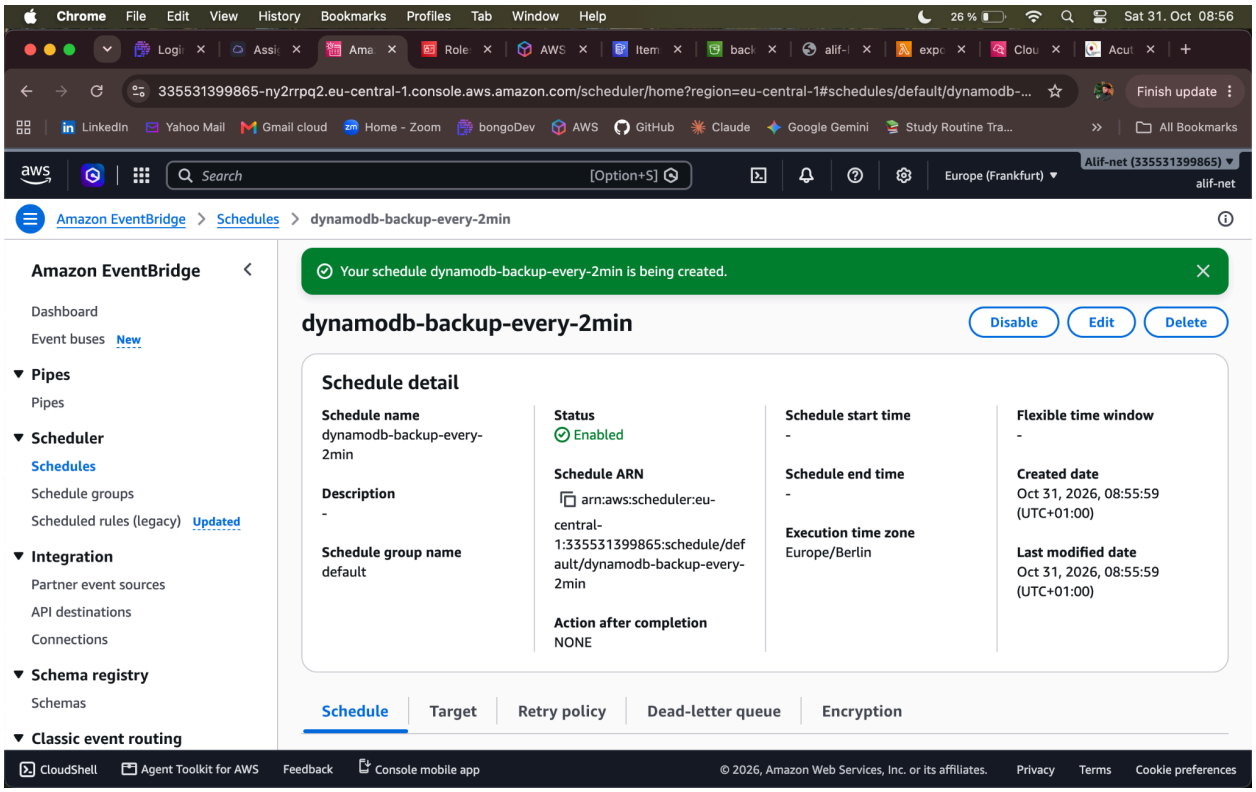

Lambda > Event Functions > import-jsonplaceholder-users
EXPLORER
IMPORT-JSONPLACEHOLDER-USERS
lambda_function.py
DEPLOY ✓ Current
Deploy (⇧⌘U)
Test (⇧⌘I)
TEST EVENTS [NONE SELECTED]
Create new test event
ENVIRONMENT VARIABLES
lambda_function.py
import json
import os
import urllib.request
import boto3
URL = "https://jsonplaceholder.typicode.com/users"
dynamodb = boto3.resource("dynamodb")
def lambda_handler(event, context):
table_name = os.environ["TABLE_NAME"]
table = dynamodb.Table(table_name)
print(f"Fetching users from {API_URL}")
PROBLEMS
OUTPUT
CODE REFERENCE LOG
TERMINAL
Execution Results
Status: Succeeded
Test Event Name: (unsaved) ImportUserTest
Response:
{
"statusCode": 200,
"body": "{\"table\": \"jsonplaceholder-users\", \"saved_count\": 10}"
}
The area below shows the last 4 KB of the execution log.
Lambda Deployed
Amazon Q
Ln 33, Col 1 (70 selected)
Spaces: 4
UTF-8
LF
Python
Lambda
Layout: U.S.

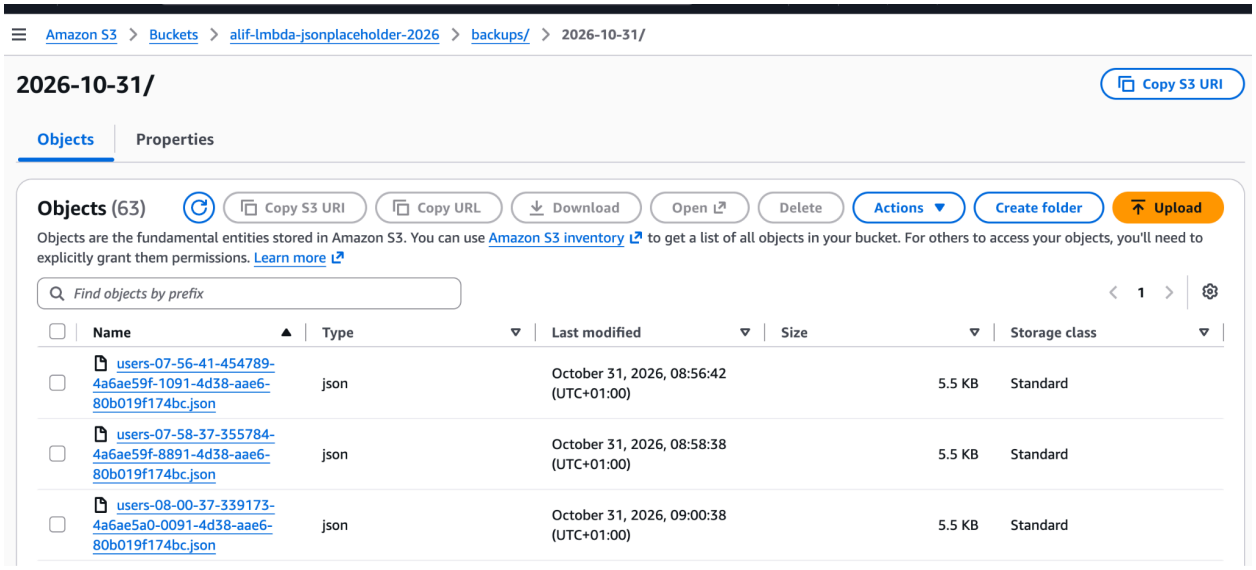

CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Metrics
Omni New
Observability Workshops
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
1m
1h
UTC timezone
Display
Timestamp
Message
2026-10-30T11:32:27.308Z INIT_START Runtime Version: python:3.13.mainlinev2.v81 Runtime Version ARN: arn:aws:lambda:eu-central-1::r…
2026-10-30T11:32:27.829Z START RequestId: bf075cf8-e234-42b7-823f-b7ed6e35c4d4 Version: $LATEST
2026-10-30T11:32:27.842Z Fetching users from https://jsonplaceholder.typicode.com/users
2026-10-30T11:32:28.446Z Saved user 1: Leanne Graham
2026-10-30T11:32:28.482Z Saved user 2: Ervin Howell
2026-10-30T11:32:28.521Z Saved user 3: Clementine Bauch
2026-10-30T11:32:28.561Z Saved user 4: Patricia Lebsack
2026-10-30T11:32:28.601Z Saved user 5: Chelsey Dietrich
2026-10-30T11:32:28.621Z Saved user 6: Mrs. Dennis Schulist
2026-10-30T11:32:28.661Z Saved user 7: Kurtis Weissnat
2026-10-30T11:32:28.701Z Saved user 8: Nicholas Runolfsdottir V
2026-10-30T11:32:28.721Z Saved user 9: Glenna Reichert
2026-10-30T11:32:28.761Z Saved user 10: Clementina DuBuque
2026-10-30T11:32:28.761Z SUCCESS: Saved 10 users to DynamoDB table jsonplaceholder-users
2026-10-30T11:32:28.783Z END RequestId: bf075cf8-e234-42b7-823f-b7ed6e35c4d4
Back to top

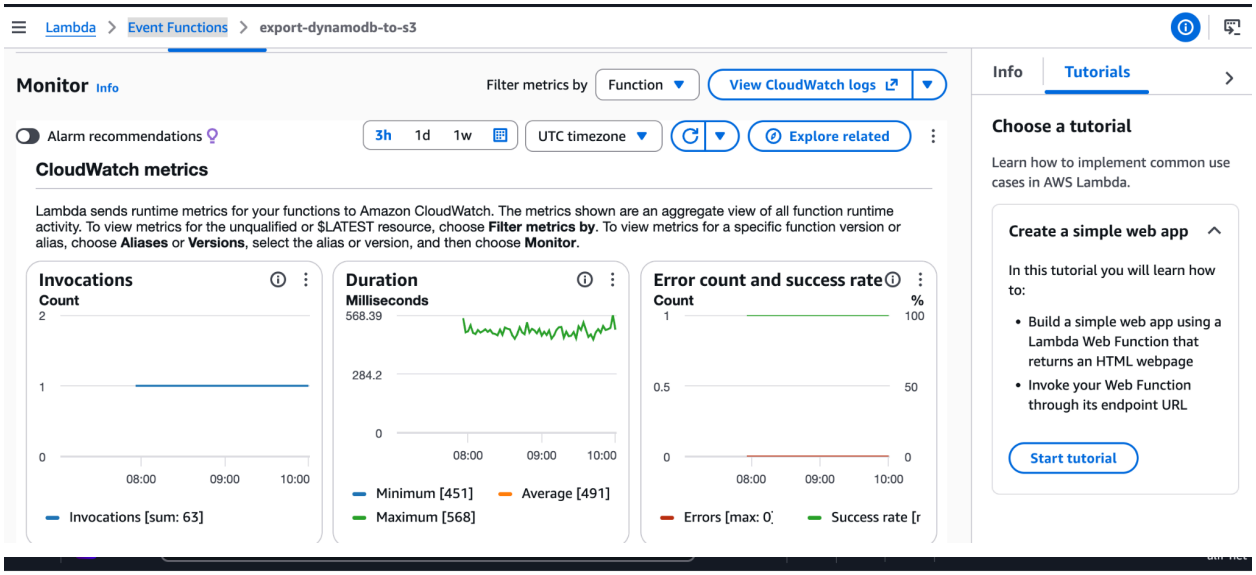

DynamoDB > Explore items > jsonplaceholder-users
DynamoDB
Dashboard
Tables
Explore items
PartiQL editor
Backups
Exports to S3
Imports from S3
Integrations
Reserved capacity
Settings
DAX
Clusters
Subnet groups
Parameter groups
Events
Tables (1)
Filter by tag
Any tag key
Filter by tag value
Any tag value
Find tables
1
jsonplaceholder-users
jsonplaceholder-users
Autopreview
View table details
Scan, query or search items Info
Scan
Query
Search
Select a table or index
Table - jsonplaceholder-users
Select attribute projection
All attributes
Filters - optional
Run
Reset
Completed · Items returned: 10 · Items scanned: 10 · Efficiency: 100% · RCUs consumed: 2

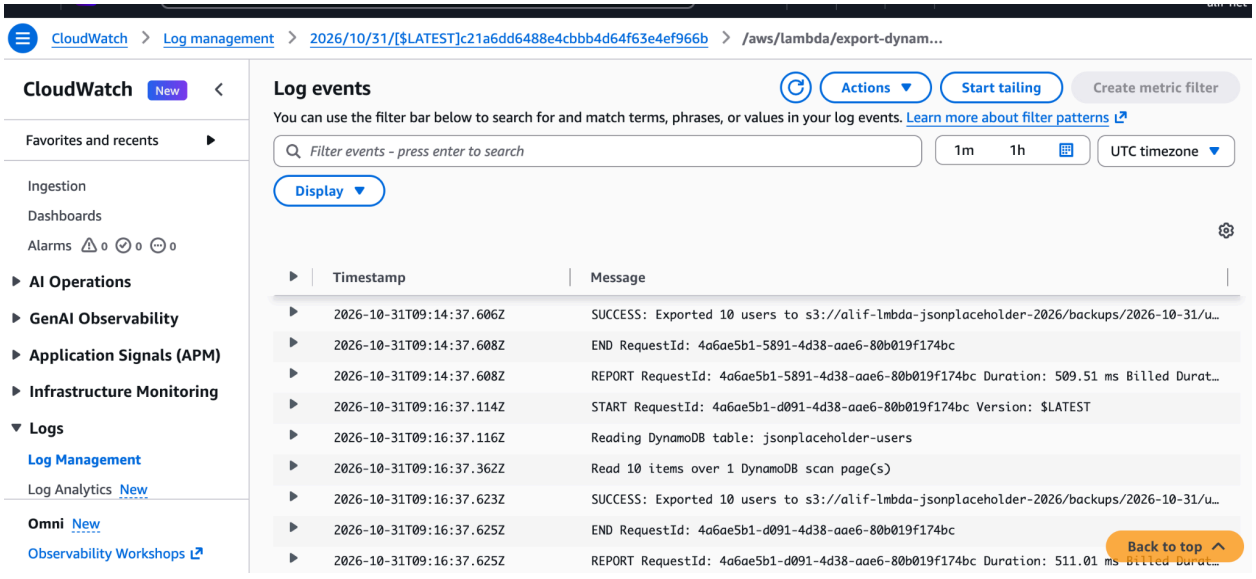

Lambda > Event Functions > export-dynamodb-to-s3
export-dynamodb-to-s3
Throttle
Copy ARN
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
export-dynamodb-to-s3
Layers (0)
Add trigger
Add destination
Function ARN
arn:aws:lambda:eu-central-1:335531399865:function:export-dynamodb-to-s3
Description
-
Last modified
22 minutes ago
Code
Test
Monitor
Configuration
Aliases
Versions
Executing function: succeeded (logs)
Details
Test event Info
Delete
CloudWatch Logs Live Tail
Save
Test
To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app using a Lambda Web Function that returns an HTML webpage
Invoke your Web Function through its endpoint URL
Start tutorial


Search
[Option+S]
Ask Amazon Q
Europe (Frankfurt)
alif-net
Lambda > Event Functions > export-dynamodb-to-s3
Executing function: succeeded (logs)
Details
Response
Format JSON
{
"statusCode": 200,
"body": "{\"exported_count\": 10, \"s3_uri\": \"s3://alif-lmbda-jsonplaceholder-2026/backups/2026-10-30/users-21-02-46-293364-77033251-1390-4733-bec1-eac63e513ee4.json\", \"scan_pages\": 1}"
}
Summary
Code SHA-256
kAiGXIK+2CCJ0TgCuUp+OxA3OwDvQSZNKTrtsaUC9UA=
Execution time
41 seconds ago
Function version
$LATEST
Request ID
77033251-1390-4733-bec1-eac63e513ee4
Duration
587.85 ms
Billed duration
1180 ms
Resources configured
128 MB
Max memory used
96 MB
Init duration
591.49 ms
Log output
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app using a Lambda Web Function that returns an HTML webpage
Invoke your Web Function through its endpoint URL
Start tutorial


Chrome File Edit View History Bookmarks Profiles Tab Window Help
Sat 31. Oct 08:56
335531399865-ny2rrpq2.eu-central-1.console.aws.amazon.com/scheduler/home?region=eu-central-1#schedules/default/dynamodb-...
Finish update
LinkedIn
Yahoo Mail
Gmail cloud
Home - Zoom
bongoDev
AWS
GitHub
Claude
Google Gemini
Study Routine Tra...
All Bookmarks
Search
[Option+S]
Europe (Frankfurt)
Alif-net (335531399865)
alif-net
Amazon EventBridge > Schedules > dynamodb-backup-every-2min
Amazon EventBridge
Dashboard
Event buses New
Pipes
Pipes
Scheduler
Schedules
Schedule groups
Scheduled rules (legacy) Updated
Integration
Partner event sources
API destinations
Connections
Schema registry
Schemas
Classic event routing
Your schedule dynamodb-backup-every-2min is being created.
dynamodb-backup-every-2min
Disable
Edit
Delete
Schedule detail
Schedule name
dynamodb-backup-every-2min
Status
Enabled
Schedule start time
-
Flexible time window
-
Description
-
Schedule ARN
arn:aws:scheduler:eu-central-1:335531399865:schedule/default/dynamodb-backup-every-2min
Schedule end time
-
Created date
Oct 31, 2026, 08:55:59 (UTC+01:00)
Schedule group name
default
Execution time zone
Europe/Berlin
Last modified date
Oct 31, 2026, 08:55:59 (UTC+01:00)
Action after completion
NONE
Schedule
Target
Retry policy
Dead-letter queue
Encryption
CloudShell
Agent Toolkit for AWS
Feedback
Console mobile app
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Amazon S3 > Buckets > alif-lmbda-jsonplaceholder-2026 > backups/ > 2026-10-31/
2026-10-31/
Copy S3 URI
Objects
Properties
Objects (63)
Copy S3 URI
Copy URL
Download
Open
Delete
Actions
Create folder
Upload
Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more
Find objects by prefix
Name
Type
Last modified
Size
Storage class
users-07-56-41-454789-4a6ae59f-1091-4d38-aae6-80b019f174bc.json
json
October 31, 2026, 08:56:42 (UTC+01:00)
5.5 KB
Standard
users-07-58-37-355784-4a6ae59f-8891-4d38-aae6-80b019f174bc.json
json
October 31, 2026, 08:58:38 (UTC+01:00)
5.5 KB
Standard
users-08-00-37-339173-4a6ae5a0-0091-4d38-aae6-80b019f174bc.json
json
October 31, 2026, 09:00:38 (UTC+01:00)
5.5 KB
Standard


Lambda > Event Functions > export-dynamodb-to-s3
Monitor Info
Filter metrics by
Function
View CloudWatch logs
Alarm recommendations
3h
1d
1w
UTC timezone
Explore related
CloudWatch metrics
Lambda sends runtime metrics for your functions to Amazon CloudWatch. The metrics shown are an aggregate view of all function runtime activity. To view metrics for the unqualified or $LATEST resource, choose Filter metrics by. To view metrics for a specific function version or alias, choose Aliases or Versions, select the alias or version, and then choose Monitor.
Invocations
Count
Invocations [sum: 63]
Duration
Milliseconds
568.39
284.2
Minimum [451]
Average [491]
Maximum [568]
Error count and success rate
Count
%
Errors [max: 0]
Success rate [r
Info
Tutorials
Choose a tutorial
Learn how to implement common use cases in AWS Lambda.
Create a simple web app
In this tutorial you will learn how to:
Build a simple web app using a Lambda Web Function that returns an HTML webpage
Invoke your Web Function through its endpoint URL
Start tutorial


CloudWatch > Log management > 2026/10/31/[$LATEST]c21a6dd6488e4cbbb4d64f63e4ef966b > /aws/lambda/export-dynam...
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management
Log Analytics New
Omni New
Observability Workshops
Log events
Actions
Start tailing
Create metric filter
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Filter events - press enter to search
1m
1h
UTC timezone
Display
Timestamp
Message
2026-10-31T09:14:37.606Z SUCCESS: Exported 10 users to s3://alif-lmbda-jsonplaceholder-2026/backups/2026-10-31/u…
2026-10-31T09:14:37.608Z END RequestId: 4a6ae5b1-5891-4d38-aae6-80b019f174bc
2026-10-31T09:14:37.608Z REPORT RequestId: 4a6ae5b1-5891-4d38-aae6-80b019f174bc Duration: 509.51 ms Billed Durat…
2026-10-31T09:16:37.114Z START RequestId: 4a6ae5b1-d091-4d38-aae6-80b019f174bc Version: $LATEST
2026-10-31T09:16:37.116Z Reading DynamoDB table: jsonplaceholder-users
2026-10-31T09:16:37.362Z Read 10 items over 1 DynamoDB scan page(s)
2026-10-31T09:16:37.623Z SUCCESS: Exported 10 users to s3://alif-lmbda-jsonplaceholder-2026/backups/2026-10-31/u…
2026-10-31T09:16:37.625Z END RequestId: 4a6ae5b1-d091-4d38-aae6-80b019f174bc
2026-10-31T09:16:37.625Z REPORT RequestId: 4a6ae5b1-d091-4d38-aae6-80b019f174bc Duration: 511.01 ms
Back to top